

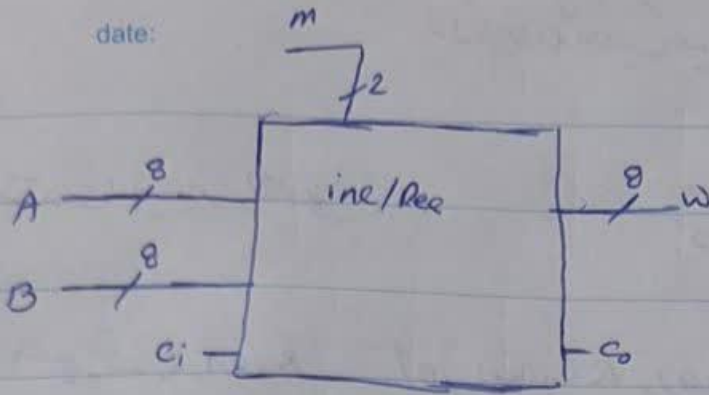
In the name of god

Morteza mohammadi motlagh

810102511

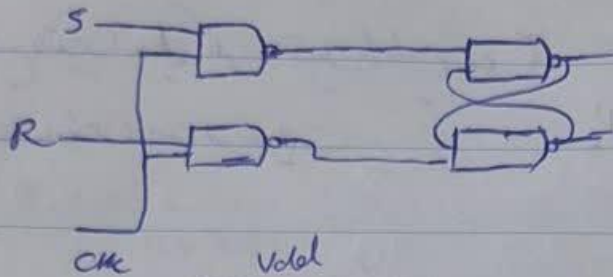
date:

subject:



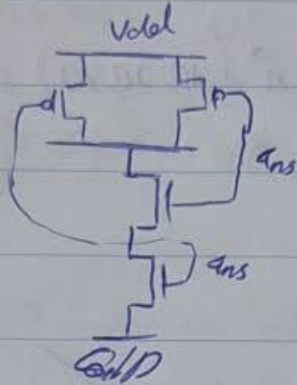
No Delay

(Q₁)

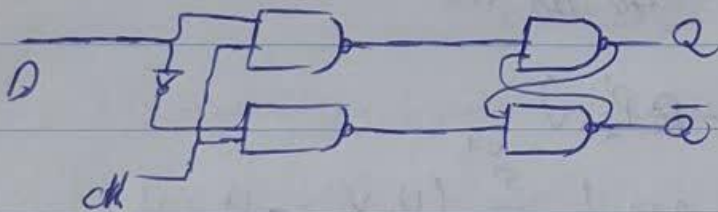


SR latch delay = 16ns

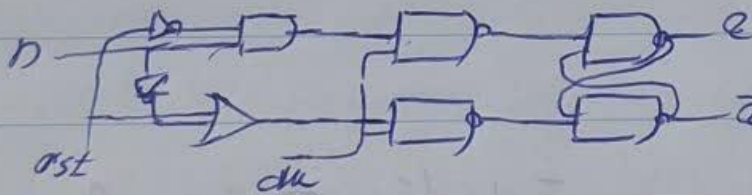
(Q₃)



NAND delay = 8ns



D latch delay = 6ns (Q₄)



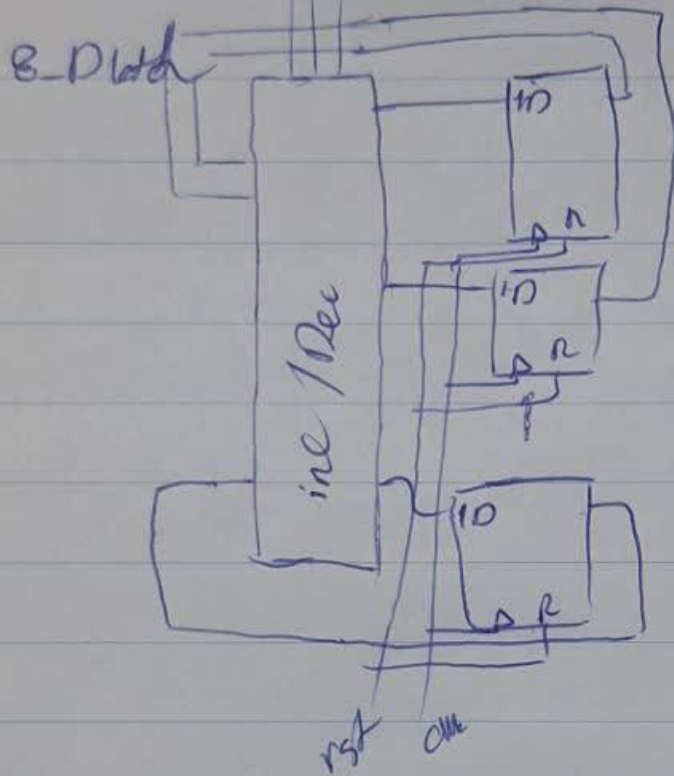
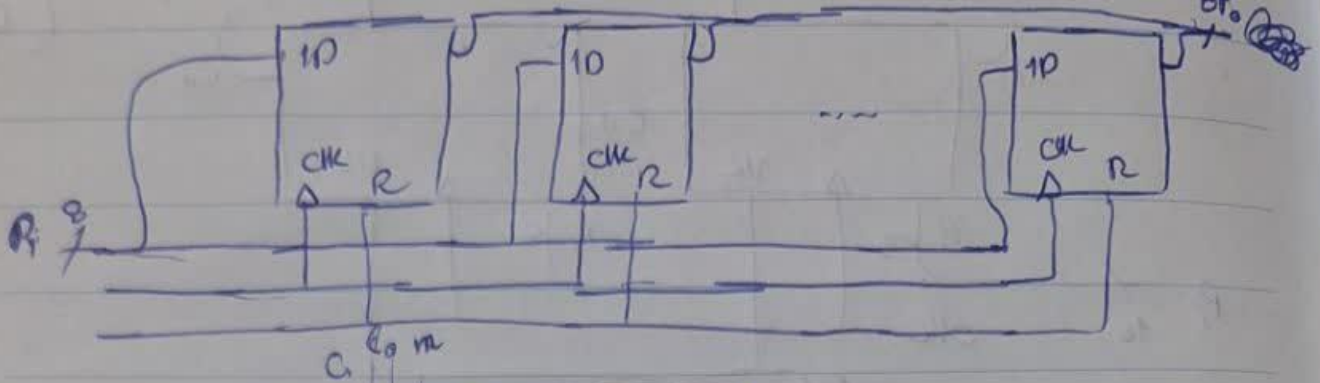
D latch with rst

delay = ~~34~~ 39ns

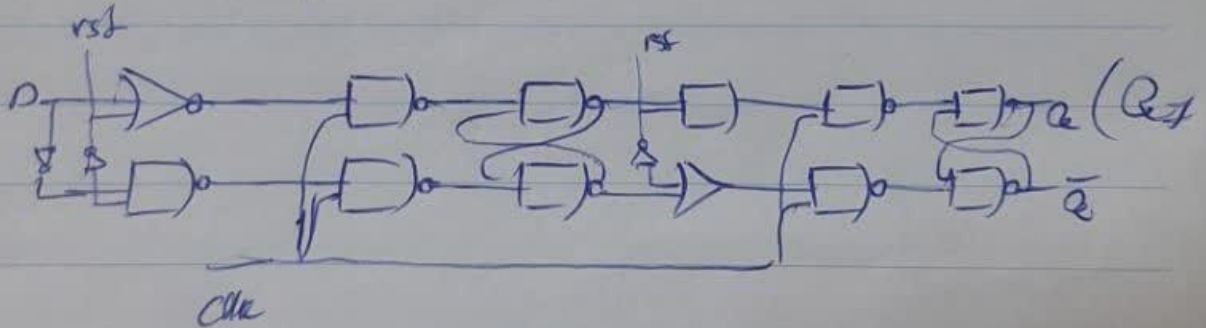
date: / /

subject: Digital Electronics

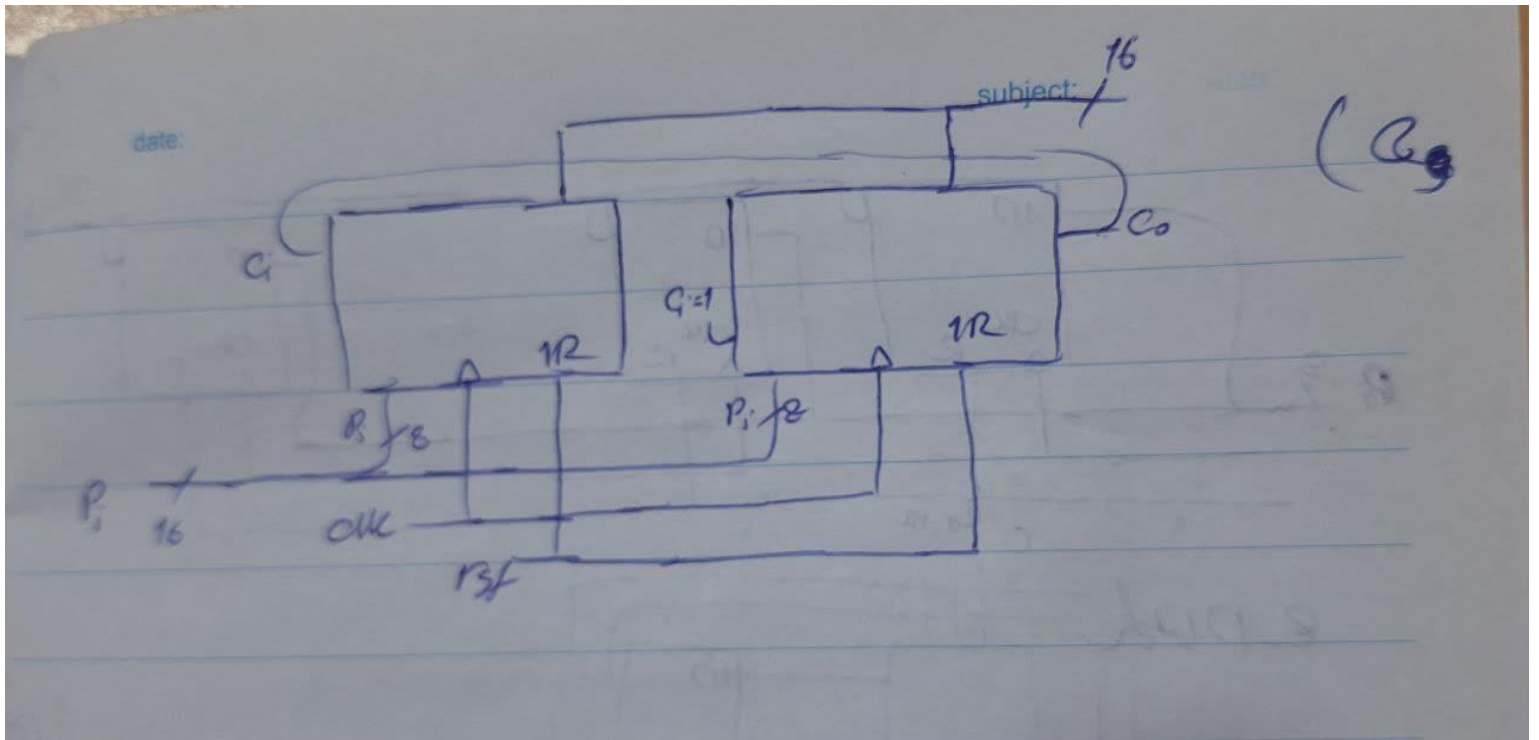
Q.1



Delay = 34 ns + (inc/Dec) delay



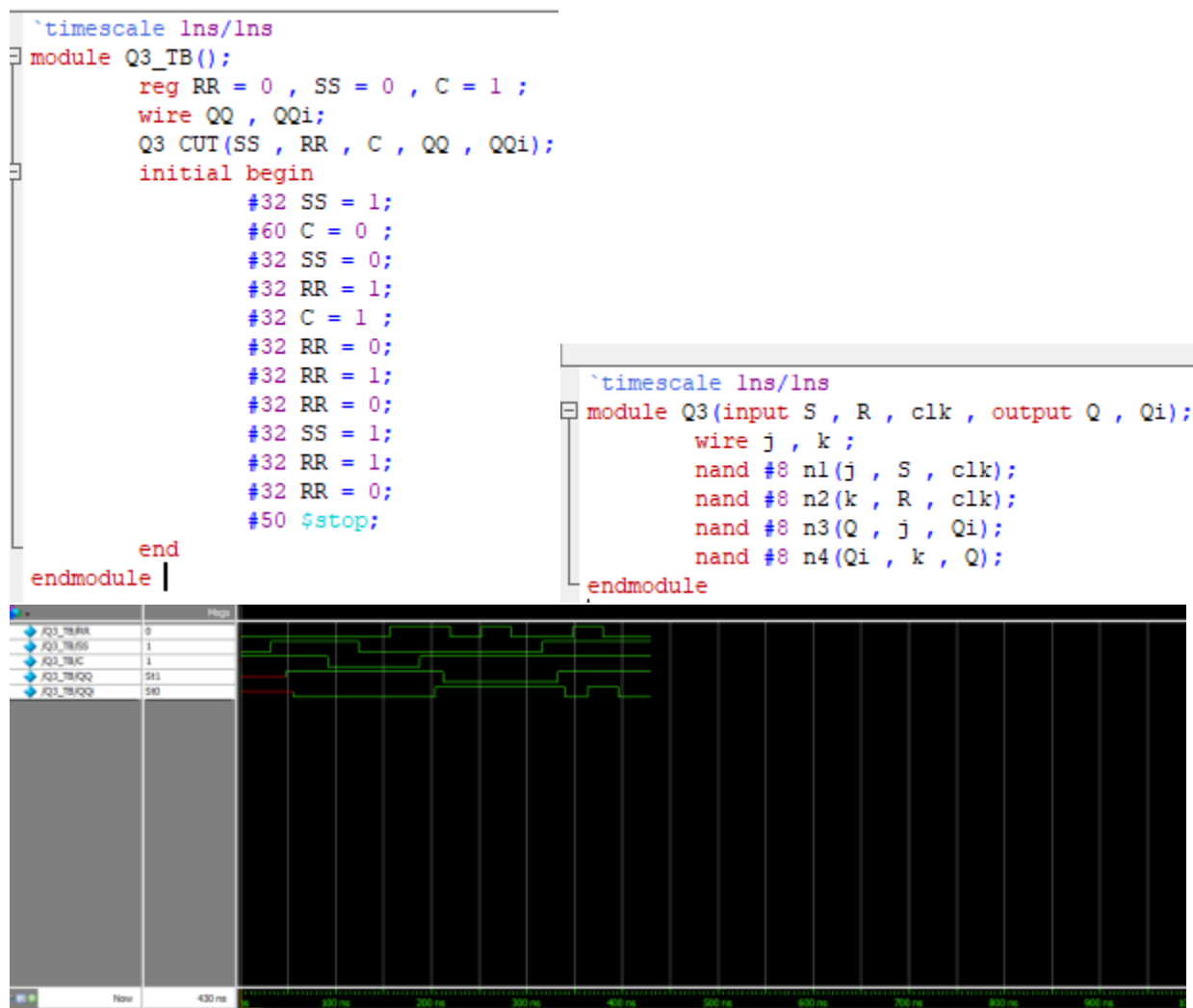
2-bit flip flop delay = $2 \times \frac{34}{2} = 68 \text{ ns}$



codes and tsetbenchs and waves

Q1)

Q3)



Q4)

```

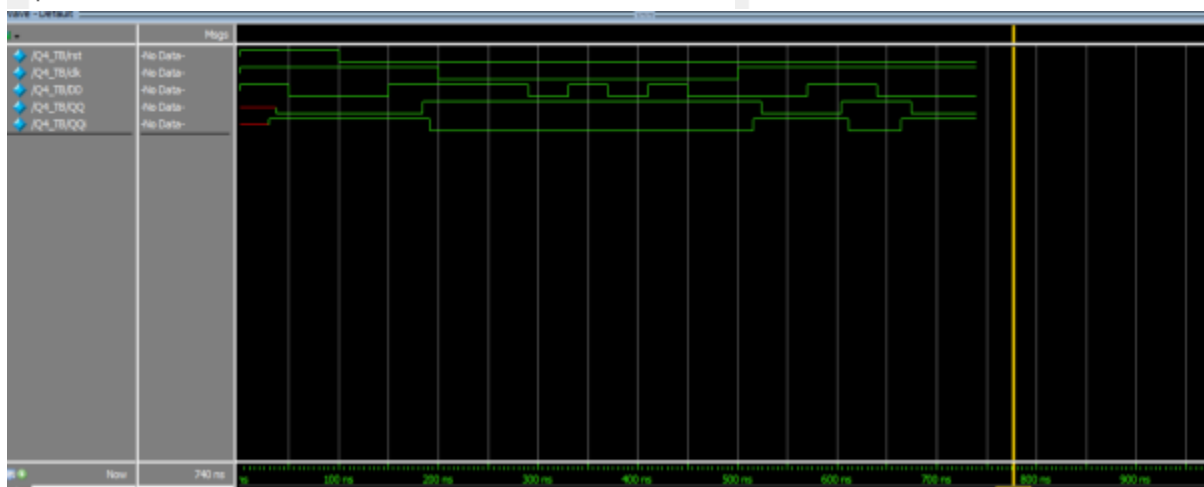
`timescale 1ns/1ns
module Q4 (input D , clk , rst , output Q , Qi);
    wire D_inv , s , r ;
    not #6 i1(D_inv , D);
    not #6 i2(rst_inv , rst);
    nor #12 i3(s , rst , D_inv);
    nand #8 i4(r , D , rst_inv);
    Q3 Dlatch(s , r , clk , Q , Qi);
endmodule

```

```

`timescale 1ns/1ns
module Q4_TB ();
    reg rst = 1 , clk = 1 , DD = 1 ;
    wire QQ , QQi;
    Q4 CUT1(DD , clk , rst , QQ , QQi);
    initial begin
        #50 DD = 0;
        #50 rst = 0;
        #50 DD = 1 ;
        #50 clk = 0;
        #50 repeat (5) #40 DD = ~DD;
        #50 clk = 1 ;
        #70 DD = ~DD;
        #70 DD = ~DD;
        #100 $stop ;
    end
endmodule

```



Q5)

```

`timescale 1ns/1ns
module Q5 (input [7:0] Pi, input clk, rst, output [7:0] Po );

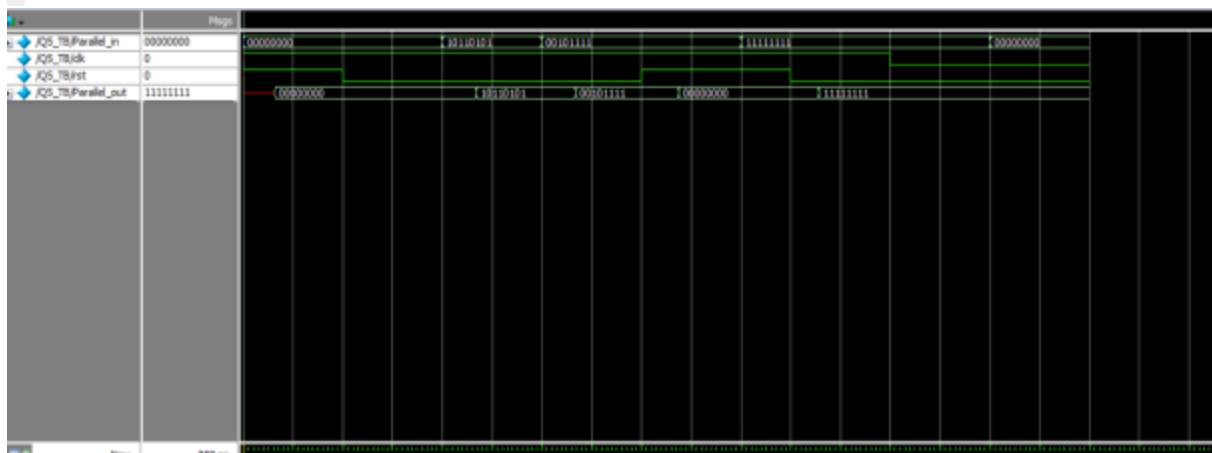
    genvar i;
    generate
        for (i = 0; i < 8; i = i + 1) begin : gen_ff
            Q4 ff_inst (.D(Pi[i]), .clk(clk), .rst(rst), .Q(Po[i]));
        end
    endgenerate
endmodule

```

```

`timescale 1ns/1ns
module Q5_TB();
    reg [7:0] Parallel_in ;
    reg clk = 1 , rst = 1;
    wire [7:0] Parallel_out;
    Q5 CUT2(Parallel_in , clk , rst , Parallel_out);
    initial begin
        Parallel_in = 8'b0;
        #100 rst = 0;
        #100 Parallel_in = 8'b10110101;
        #100 Parallel_in = 8'b00101111;
        #100 rst = 1;
        #100 Parallel_in = 8'b11111111;
        #50 rst = 0;
        #100 clk = 0;
        #100 Parallel_in = 8'b0;
        #100 $stop;
    end
endmodule

```



Q6)

```

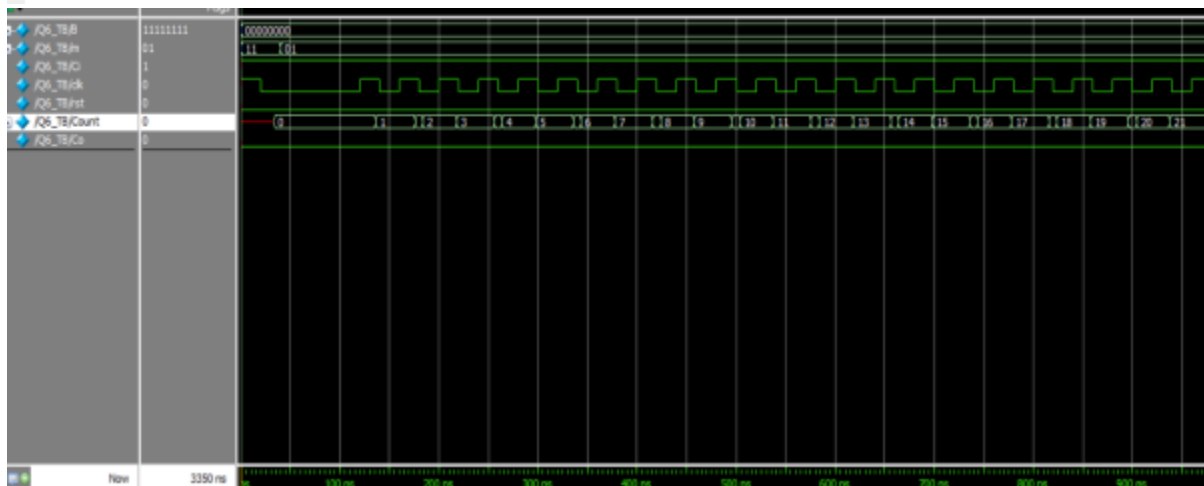
`timescale 1ns/1ns
module Q6(input [7:0] B, input [1:0] m, input Ci, input clk, input rst, output [7:0] Count, output Co);
    wire [7:0] W;
    Q1_aa m1(Count , B , m , Ci , W, Co);
    Q5 m2(W , clk , rst , Count);
endmodule

```

```

`timescale 1ns/1ns
module Q6_TB();
    reg [7:0] B;
    reg [1:0] m;
    reg Ci , clk , rst;
    wire [7:0] Count;
    wire Co;
    Q6 CUT56(B , m , Ci , clk , rst , Count , Co);
    initial begin
        B = 8'b0;
        m = 2'b11;
        clk = 1;
        Ci = 1;
        rst = 0;
        #20 clk = 0 ;
        #20 m = 2'b01;
        #60 ;repeat (60) #20 clk = ~clk;
        #60 rst = 1;
        #30 clk = 1;
        #30 rst = 0;repeat (50) #20 clk = ~clk;
        #25 Ci = 0;
        #60 m = 2'b10;repeat (30) #20 clk = ~clk;
        #20 clk = 0;
        #60 B = 8'b11111111;
        #20 m = 2'b11;
        #10 clk = 1;
        #20 clk = 0;
        #5 Ci = 1;
        #20 m = 2'b01;
        #10 clk = 1;
        #20 clk = 0;
        #60 $stop;
    end
endmodule

```



```

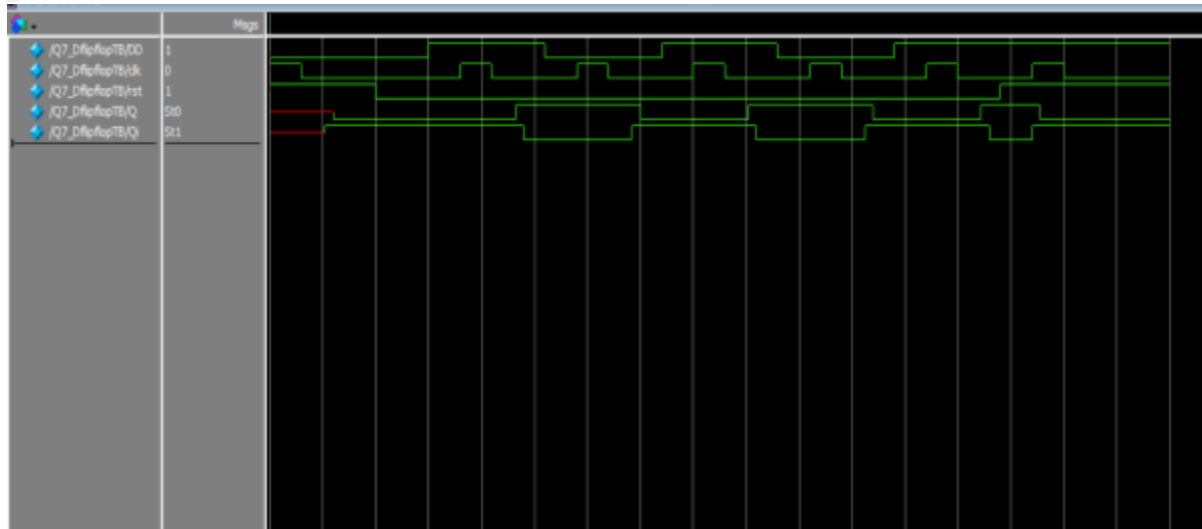
`timescale 1ns/1ns
module Q7_Dflipflop (input D , clk , rst , output Q , Qi);
    wire clk_inv , Q_1 , Qi_1 ;
    not #6 kl(clk_inv , clk);
    Q4 k2(D , clk , rst , Q_1 , Qi_1);
    Q4 k3(Q_1 , clk_inv , rst , Q , Qi);
endmodule

```

```

`timescale 1ns/1ns
module Q7_DflipflopTB();
    reg DD = 0 , clk = 1 , rst = 1;
    wire Q , Qi ;
    Q7_Dflipflop CUT4(DD , clk , rst , Q , Qi);
    initial begin
        #30 clk = 0;
        #70 rst = 0;
        #50 DD = 1;
        #30 clk = 1;
        #30 clk = 0;
        #50 DD = 0;
        #30 clk = 1;
        #30 clk = 0;
        #50 DD = 1;
        #30 clk = 1;
        #30 clk = 0;
        #50 DD = 0;
        #30 clk = 1;
        #30 clk = 0;
        #50 DD = 1;
        #30 clk = 1;
        #30 clk = 0;
        #40 rst = 1;
        #30 clk = 1;
        #30 clk = 0;
        #100 $stop;
    end
endmodule

```

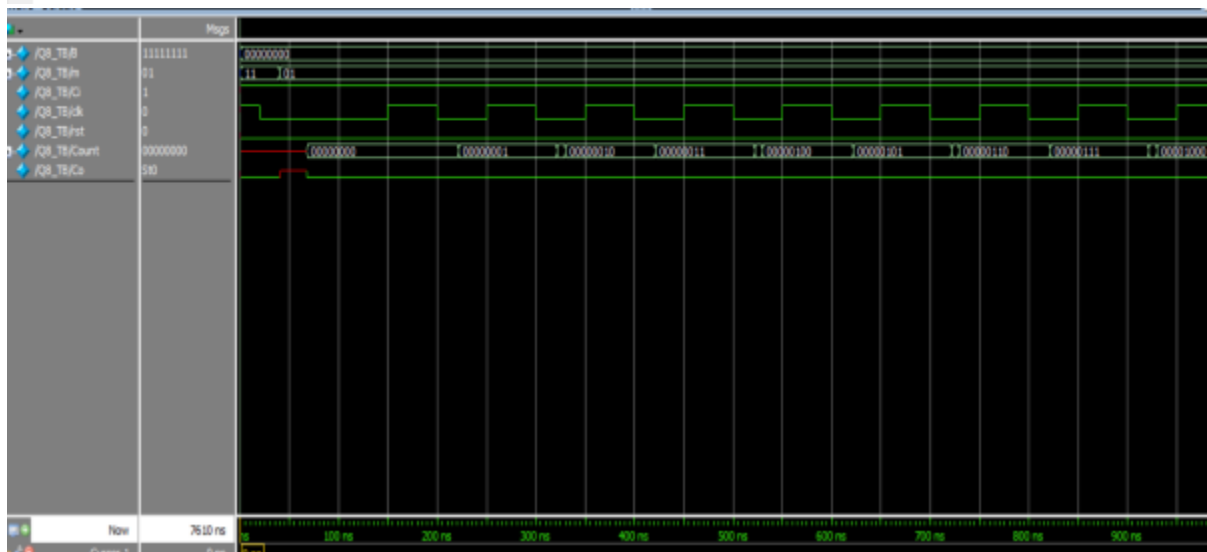


```

`timescale 1ns/1ns
module Q8(input [7:0] B, input [1:0] m, input Ci, input clk, input rst, output [7:0] Count, output Co);
    wire [7:0] W;
    Q1_aa z1(Count , B , m , Ci , W , Co);
    Q7_Register z2(W , clk , rst , Count);
endmodule

`timescale 1ns/1ns
module Q8_TB();
    reg [7:0] B;
    reg [1:0] m;
    reg Ci , clk , rst;
    wire [7:0] Count;
    wire Co;
    Q8 CUT51(B , m , Ci , clk , rst , Count , Co);
    initial begin
        B = 8'b0;
        m = 2'b11;
        clk = 1;
        Ci = 1;
        rst = 0;
        #20 clk = 0 ;
        #20 m = 2'b01;
        #60 ;repeat (60) #50 clk = ~clk;
        #60 rst = 1;
        #30 clk = 1;
        #30 rst = 0;repeat (50) #50 clk = ~clk;
        #25 Ci = 0;
        #60 m = 2'b10;repeat (30) #50 clk = ~clk;
        #20 clk = 0;
        #60 B = 8'b11111111;
        #20 m = 2'b11;
        #10 clk = 1;
        #50 clk = 0;
        #5 Ci = 1;
        #20 m = 2'b01;
        #10 clk = 1;
        #50 clk = 0;
        #60 $stop;
    end
endmodule

```



```
`timescale 1ns/1ns
module Q16(input [7:0] B_0to7, B_8to15 ,input [1:0] m ,input Ci ,input clk , rst, output [15:0] Count , output Co);
    wire Carry_out;
    Q8 j1(B_0to7 , m , Ci , clk , rst , Count[7:0] ,Carry_out);
    Q8 j2(B_8to15 , m , Carry_out , clk , rst , Count[15:8] , Co);
endmodule
```